

Automatic STREET-CUM-PORCH LIGHT

SURESH CHANDRA DWIVEDI

The street-cum-porch light integrates the functionalities of street and porch lights into a single unit, optimising space and reducing clutter. By using LEDs, it offers energy savings and allows users to customise the lighting according to their needs. The use of LM358 and timer 555 ICs provides sophisticated

control and automation features, ensuring a seamless user experience.

Fig. 1 shows the author's prototype.

Circuit and working

The circuit diagram of the street-cum-porch light is shown in Fig. 2. It is built around a 12V adaptor connected across JACK1, op-amp LM358 (IC1), timer NE555 (IC2), two BC548 transistors (T1, T2), 12V SPDT relay (RL1), and a few other components.

The NE555 timer is configured in monostable mode to generate a preset time output, thereafter, going low. One op-amp from the LM358 is used to compare voltages and the voltage developed by the LDR when light is falling on it. The NE555 timer IC2 is triggered by the output of IC1A and transistor T2 based on the light falling on the LDR. The output of the op-amp LM358 is connected to mode change switch S1. The output of the NE555 timer is also connected to mode change switch S1.

Timer IC2 with RC components decides the on time at output pin 3 to switch on the porch light via relay RL1. Here, in RC, the time-

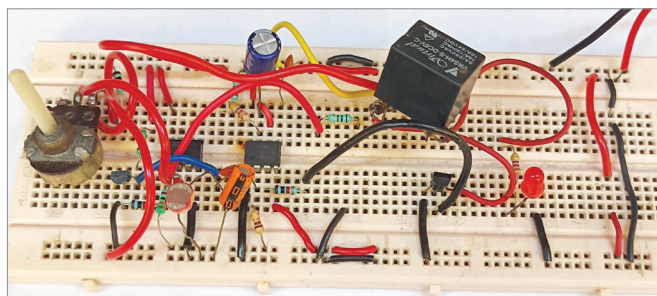


Fig. 1: Author's prototype

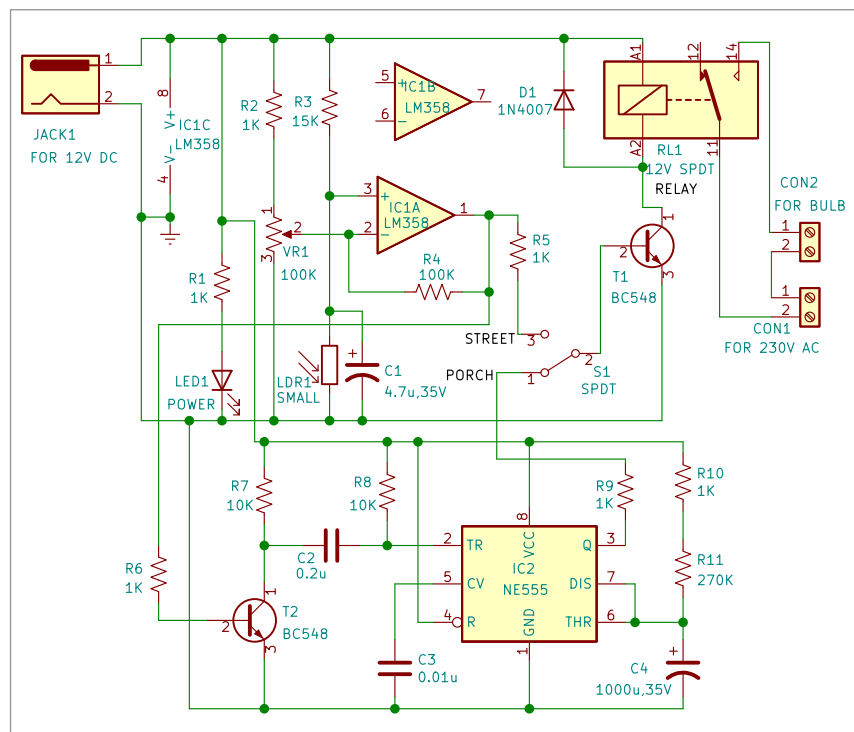


Fig. 2: Circuit diagram

PARTS LIST

Semiconductors:

- IC1 (IC1A-IC1C) - LM358 operational amplifier
- IC2 - NE555 timer
- T1, T2 - BC548 npn transistor
- D1 - 1N4007 rectifier diode
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

- R1, R2, R5, R6, R9, R10 - 1-kilo-ohm
- R3 - 15-kilo-ohm
- R4 - 100-kilo-ohm
- R7, R8 - 10-kilo-ohm
- R11 - 270-kilo-ohm
- VR1 - 120-kilo-ohm potentiometer

Capacitors:

- C1 - 4.7 μ F, 35V electrolytic
- C2 - 0.2 μ F, ceramic disk
- C3 - 0.01 μ F, ceramic disk
- C4 - 1000 μ F, 35V electrolytic

Miscellaneous:

- LDR - Small size
- JACK1 - DC socket
- CON1, CON2 - 2-pin connector
- S1 - SPDT switch
- RL1 - 12V, 1 changeover (1CO) relay
- 230V, 100-watt bulb
- 12V adaptor for power supply
- 230V AC mains power supply